



Plotting Data from a CSV Transcript

Hello everyone, welcome to bitspline digital. Today we will continue our learning journey for the course data visualisation storytelling. We'll be now looking at the libraries and commands within the Python to understand how to connect them to different data types for developing visualisations.

So generally in a data visualisation environment your underlying data is stored in standard formats like Excel, CSV or sometimes you might have to extract data directly from the web or there is it is in the HTML page right. So in all these contexts you have data lying in different set formats that has to be connected to the working environment let us say Python or Power BI. You need a specific process that has to be followed so that you import the data with integrity into your work environment and develop the visualisations from there.

So let us now look at what are the steps that you try to do in connecting such data formats to your Python work environment. Now the most common type of data that you come across is often in a CSV especially in a structured data space you have the CSV format and it can accommodate large quantities of data and each value is separated by column therefore the name comma separated values right and you get this typical dump of the data what is called in technical parlance you need to parse that into Python or your work environment and start cleaning the data and then trying to understand the patterns and then go to the visualisation part. So this is very closely related to the seven stages of data visualisations what which I have seen in the earlier classes okay.

So let us try to work out the examples of the live session wherein you will get to understand how to run the code step by step. So like in last class I am trying to run the code simultaneously so that you have the necessary templates for practising after the class. So I am just running importing the necessary libraries like NumPy, Pandas and Matplotlib and also setting the graphic style to seaborn white grade which gives a pre-established template for the graphics.

Now I have imported the data so now what I am trying to do is trying to import the comma separated values so just this is the data that you have country GDP life expectancy population and region so this is the data you have different countries you have their GDP per capita life expectancy in years right and population in million okay. Then you have the region whether it is in which continent this is the data that you have and now we can do we can import this data in two formats one is like using this directly parsing the data into the system by using data is equal to you are defining a variable called data right so and then you are saying this is separated by columns this is the entire text and you have defined the data then again you are using the pandas dot read dot CSV because each of this is now separated by a column by a comma and then you are using it is string it is a string type data and you are using the variable data

here and you are creating a data frame so let us go and try this out so this is the data I have first determined the data variable and then using that into a data frame reading that into a data frame using the command `pd.read_csv(input_string, variable_data)` and then seeing how the data is looking like so if you see the data has been read into a nice tabular format or the data frame so you have the country then you have GDP per capita life expectancy population and region now let us try to plot from this data okay so what do you do for a plot first you have to declare the figure size or set the figure right so I am using the plot setting the figure size to 10 dot 6 using `plt.figure` then

I want to see the scatter plot I want to see the relationship between GDP per capita and life expectancy so I am taking GDP per capita to be the x-axis and life expectancy to be y-axis here again the choice of parameters determined by the underlying relationship you expect richer countries to have higher life expectancy that is the hypothesis are the expectation so GDP per capita becomes X variable life expectancy becomes Y being the dependent variable and also you want to size the bubbles as per the population so you are using the size is equal to S is size is represented by S you can also write the full form size here right so then you are having the population which is scaled by a factor of 10 so that you have representation clear okay so then you have X label GDP per capita Y label life expectancy title is this then you want to have the grid lines so you are just showing trying to run the graph this is what you get now let us take away this 10 and see what will happen and you don't scale the population so this is how it will look if you don't scale the population which is also a good representation right so you can try out this so what we have be done in this step we have taken a comma separated value data and then passed it into the Python environment using this particular command `pd.read_csv` so we will look at more examples in the live session for that particular week you are free to try this from the sources that are available online you can also try from the direct documentation of Python so now let us move to connecting the data from a data frame okay

so I am generating the random numbers okay what am I doing I am generating random numbers for thousand days right I am creating three columns ABC and each call I mean there are thousand this is how the data is going to look like the next task is to understand how to read data from a data frame and then plot it what is a data frame so this quickly let us go back here this is the raw data and this has been converted into a data frame now suppose you have a data frame like this which is cleanly arranged and how do you plot from that okay so in this case what we are trying to do we are generating thousand days data let us say from two thousand first January two thousand two thousand two thousand days right and then there are random numbers in each of this column so this function is generating that particular data frame where you have the thousand rows which represent each of the day and three columns of random variables ABC which have different values on any given date okay so that's the function you are using `np.random` generator the normal distribution which is for three

values and then you are doing this okay columns ABC okay so you are generating a data frame you are using random seed here so and then let me do the run selection part only the selection is run right so then what am I doing I am determining the data frame okay DF to be sum of consecutive numbers so the over the thousand days I am developed I am determining a function called cumulative sum to simulate something called random walks the random walk is nothing but at every step it is like this let us say it is said to be the walk of a drunkard person so the drunkard person will move in any direction so it is like this so if you stand at a point and toss a coin if it's head you would take one step further

And if it is tail one you step you take one step further and after hundred or thousand trials you have to predict or to understand what is the path that is traversed by this person on an average where he would end up so this is kind of a random walk where you are the moment is completely random okay so we are determining this as like cumulative sum so what are we doing we are adding consecutive values of the countdown numbers like day one you have ABC they do will be day ones plus day to value day three will be day one plus day two plus day three like that you are trying to create a data frame so this is what I have done here so again I am running the selection then I am running the I want to see how the data frame is looking like so this is how the data frame is looking like so have consecutive values and then I am going on adding them and I am getting the cumulative sum now I want to plot the cumulative sum simply I am taking BF dot plot which plots the particular columns right particular data series so then we are doing this then we have X label which is date and then Y label is cumulative value right and then we have a PLT grid and we are showing gridded values and we are having the plot so this is how it will look like so more than how do you plot the graphic the main task here is to reading data into data frame and then defining this cumulative sum using that for plotting so you have different thousand values and then you are having the data frame okay the data frame and then you are using it for plotting right so this is how it is going to look like yeah so I'm just inquisitive to do something else here so let me take this to another just to showcase different approach or what it look like so I'm just quickly doing more code just doing this here then instead of cumulative sum right yeah and this sorry so

I just run this yeah just instead of cumulative values testing I don't know what the output will be like yes look at the difference so yeah if you don't take cumulative values you won't get the trend in this is entirely random these are three random numbers so I'm just having the random values which are just going up and down up and down up and down up and down on a daily basis but then when you trace the steps on a cumulative basis this is how you get the clean series which is a random walk this is the random movement across the three variables on a daily basis okay so just to show the difference I have done this now let us move to yeah so okay so this is fine so yeah so we have done all this.

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